Analysis of exogenous shocks using high frequency data in JGB markets

Yoshifumi Tahira, Takayuki Mizuno[A], Takaaki Ohnishi[B], and Tsutomu Watanabe[C]

Graduate School of Science and Engineering, Chuo University, Tokyo 112-8551, Japan

[A]National Institute of Informatics, Tokyo 101-8430, Japan

[B]Graduate School of Information Science and Technology, The University of Tokyo, Tokyo 113-8654, Japan

[C]Graduate School of Economics, The University of Tokyo, Tokyo 113-8654, Japan

E-mail: ytahira@phys.chuo-u.ac.jp

Abstract

We empirically investigate fluctuations in actuals and futures prices in Japanese government bonds (JGB) markets by using a tick-by-tick price data, and find some similarities between the JGB prices and other financial prices. Cumulative distribution function of JGB price fluctuations follows a power law function with an exponent of -3. Autocorrelation coefficients are around zero, namely, the prices almost like a random walk. It is observed that a long memory property in the volatility of price changes. We also investigate exogenous shocks in the JGB markets using about 30 million news articles. Our results show that large volatilities can be often explained by the flow of news.

Keyword: Japanese government bonds, Financial news, Exogenous shock

We first show the basic properties of JGB price fluctuations by analyzing tick-by-tick data in JGB actuals and futures markets from 2010 to 2014. Second, we investigate exogenous shocks in the markets using raw texts of more than 24 million news records provided by Thompson Reuters.

We find that the price change distribution for JGB prices is close to a log normal distribution, rather than a power law distribution with an exponent of -3. We also find that the estimate of the Hurst exponent is close to 0.5 in long time scales, and that there is almost no autocorrelation in price changes, suggesting that the price process is close to a random walk. There is long memory in the market activities are measured by the volatility, volume, and number of transactions. These are quite similar to the property reported by previous studies about other financial markets.

Next, we focus on exogenous shocks in JGB markets. We observe a good correspondence between the time evolution of the market activities and the time evolution of news volume. This correspondence is showed in Fig. 1, which expresses the time evolution of the daily volatility and the evolution of the daily volume of news, measured as the number of news words in articles that include "JGB", "JGBs" and "Japanese government bonds". The cross correlation is about 0.4.

In this conference, we also focus on the balance between market order volume and limit order volume. In May 2013, JGB price dived after the crisis of U.S. bonds. At that time, the volume balance collapsed greatly.

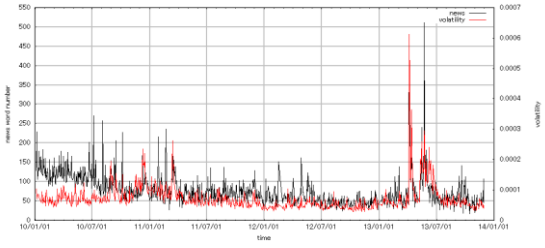


Fig. 1. Time series of price volatility and number of news words